Name: ______________________

Code: ______________________

Numerical methods

Computer Science BSc

May 25, 2023

Test 2

Solve the below problems! Write your solution of each problem on a separate page.

1. We are using the iteration $x_{k+1} = \dfrac{\cos(x_k^2) + 1}{4}$ to approximate the solution of the equation $4x - \cos(x^2) - 1 = 0$ in the interval $[0, \pi/2]$.

(a) Prove convergence!

(b) Give an error estimation!

(6+2 points)

2. Consider the function $f(x) = \ln(x) + x^2 - 2$.

(a) Show that the interval $[1, 3]$ contains a root!

(b) Write Newton's method to approximate the root!

(c) Verify that the conditions of the monotone convergence theorem hold! How to choose the starting point?

(2+2+4 points)

3. Interpolate the function $f(x) = \sqrt{x+1} - x$ on the base points $-1, 0, 3, 8$.

(a) Give the Newton form of the interpolating polynomial! (No need to expand.)

(b) What base points should we choose for a quadratic interpolation on the interval $[0, 8]$ to minimize the error?

(c) Write the error estimation in the latter case!

(7+3+4 points)

4. Find the best line and parabola fitted on the points (x_i, y_i) using least squares approximation, given

x_i	-1	0	1	2
y_i	1	-1	1	1

(4+6 points)

5. Consider the integral $\displaystyle\int_0^1 \cos(\pi x^2)\, \mathrm{d}x$.

(a) Approximate its value using midpoint, trapezoid and Simpson's formula!

(b) Estimate the error of the trapezoid formula!

(c) Find m in order to get an approximation for the integral using the composite trapezoid rule T_m within the error 10^{-2}.

(3+3+4 points)

Good work!